MILITARY SPECIFICATION

# FREQUENCY MULTIPLIERS: CV-1076/USM-129 THROUGH CV-1080/USM-129

## 1. SCOPE

**1.1 Scope.** This specification covers a series of frequency multipliers in the frequency range of 4.0 kilomegacycles per second (kmcps) to 40.0 kmcps. These frequency multipliers form part of Microwave Calibration Test Sets AN/USM-129, AN/USM-129A and AN/USM-129B.

**1.2 Classification.** The frequency multipliers covering the frequency range of 4.0 to 40.0 kmcps shall be designated by the nomenclature given in table I.

TABLE I. *Frequency multiplier-designation*

| Frequency range (kmcps) | Designator | Ordnance part No. |
|---|---|---|
| 4.0 to 8.2 | CV-1076/USM-129 | 10519268 |
| 8.2 to 12.4 | CV-1080/USM-129 | 10519230 |
| 12.4 to 18.0 | CV-1079/USM-129 | 10519231 |
| 18.0 to 26.5 | CV-1078/USM-129 | 10519232 |
| 26.5 to 40.0 | CV-1077/USM-129 | 10519233 |

## 2. APPLICABLE DOCUMENTS

**2.1** The following documents of the issue in effect on date of invitations for bids, form a part of this specification to the extent specified herein.

### SPECIFICATION

MILITARY

MIL-I-45208 —Inspection Requirements; General Specification For.

MIL-U-46351 —Universal Microwave Calibrators (UMC).

### DRAWINGS

ORDNANCE CORPS

B10519230 —Frequency Multiplier, 8.2–12.4 kmcps.

B10519231 —Frequency Multiplier, 12.4–18.0 kmcps, and Frequency Multiplier, 18.0–26.5 kmcps.

C10519233 —Frequency Multiplier, 26.5–40.0 kmcps.

C10519268 —Frequency Multiplier, Coaxial.

A10519341 —Calorimeter.

A10519351 —Calorimeter.

A10528591 —Power Meter, Amplifier, Calorimetric.

### PUBLICATIONS

CONSOLIDATED FREIGHT CLASSIFICATION RULES.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Initial Production Sample.** Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

**3.2 Materials.** Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

**3.3 Construction.** The frequency multipliers shall be manufactured in accordance with the applicable drawing (see 2.1), as required, and all drawings referenced thereon. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired electrical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities.

**3.4 Electrical.**

3.4.1 *Output Frequency Range.* The harmonic output frequency range shall be from 4.0 to 40.0 kmcps as shown in table II.

TABLE II. *Harmonic output frequency range (kmcps)*

| Ordnance part No. | Frequency range |
|---|---|
| 10519230 | 4.0— 8.2 |
| 10519268 | 8.2—12.4 |
| 10519231 | 12.4—18.0 |
| 10519232 | 18.0—26.5 |
| 10519233 | 26.5—40.0 |

3.4.2 *Output Power.* The harmonic power output in decibels below one milliwatt (—dbm) for an input frequency of 4.0 kmcps and an input power of 100 milliwatts shall be as shown in table III.

TABLE III. *Harmonic output power*

| Ordnance part No. | Input Frequency (kmcps) | Harmonic output frequency (kmcps) | Input power milliwatts | Harmonic output power (dbm) |
|---|---|---|---|---|
| 10519268 | 4.0 | 8.0 | 100 | > 0 |
| 10519230 | 4.0 | 12.0 | 100 | >—10 |
| 10519231 | 4.0 | 16.0 | 100 | >—20 |
| 10519232 | 4.0 | 24.0 | 100 | >—30 |
| 10519233 | 4.0 | 32.0 | 100 | >—40 |

**3.5 Environmental.**

3.5.1 *Operating.*

3.5.1.1 *Temperature.* Prior to, during and after being exposed and thermally stabilized at ambient temperatures of plus 32° and plus 110° F., the frequency multiplier shall meet the requirements of 3.4.

3.5.1.2 *Humidity.* Prior to, during and after being subjected to the procedure specified in 4.7.1.2, the frequency multiplier shall meet the requirements of 3.4.

3.5.1.3 *Altitude.* The frequency multiplier shall be capable of operating at a reduced pressure of 20.58 inches of mercury (equal to an altitude of 10,000 feet), for four hours.

3.5.2 *Non-operating.*

3.5.2.1 *Temperature.* Prior to and after being subjected to the procedure specified in 4.7.2.1, the frequency multiplier shall meet the requirements of 3.4.

3.5.2.2 *Altitude.* Prior to and after being subjected to the procedure specified in 4.7.2.2, the frequency multiplier shall meet the requirements of 3.4.

3.5.2.3 *Bench Handling.* After being subjected to the procedure specified in 4.7.2.3, the frequency multiplier shall show no physical damage and shall meet the requirements of 3.4.

3.5.2.4 *Vibration.* After being subjected to the procedure specified in 4.7.2.4, the frequency multiplier shall show no physical damage and shall meet the requirements of 3.4.

3.5.2.5 *Shock.* After being subjected to the procedure specified in 4.7.2.5, the frequency multiplier shall show no physical damage and shall meet the requirements of 3.4.

**3.6 Workmanship.** The equipment, including all parts and accessories shall be manufactured and finished in a thoroughly workmanlike manner and in accordance with good commercial practice.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibilities.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.2 General Requirements.** Inspection shall be in accordance with Specification MIL–I–45208.

**4.3 Initial Production Sample.** Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

**4.3.1** *Production Sample Failure.* Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

**4.4 Material.** Control of material shall be in accordance with test data requirements of Specification MIL–I–45208.

**4.5 Inspection of Components.** The contractor shall ascertain that prior to assembly all components procured under separate specifications or drawings, have been inspected, tested and accepted in accordance with their respective specifications or drawings.

**4.6 Electrical.**

**4.6.1** *Output Power.* At the input frequency given in 3.4.2, the input power from a suitable signal source shall be fed through a tuner into the frequency multiplier. The output signal of the frequency multiplier shall be fed to a suitable receiver. The proper harmonic output frequency shall be selected from table III. The proper bias voltage and current shall be applied to the varactor of the frequency multiplier. The output signal of the frequency multiplier as displayed on the oscilloscope of the receiver shall be maximized by means of the tuner and the tuning mechanism of the frequency multiplier. The magnitude of the display shall be recorded. A calibrated 1 milliwatt signal as monitored by a dry calorimeter (drawing A10519341 or A10519351, as required), calorimetric power meter (drawing A10528591) or approved equivalents shall be applied to the receiver. Using a precision attenuator with an accuracy of at least ±0.4 db, the 1 milliwatt signal shall be attenuated until the display is equal in amplitude to that of the frequency multiplier. The amount of attenuation (db) as indicated by the precision attenuator shall comply with the values for harmonic output power as given in table III.

**4.7 Environmental.**

**4.7.1** *Operating Tests.*

**4.7.1.1** *Temperature.* When tested while exposed and thermally stabilized at ambient temperatures of plus 32° and plus 110° F., the frequency multiplier shall conform to the requirements of 3.4.

**4.7.1.2** *Humidity.* After having been exposed to an atmosphere of 95 percent relative humidity at plus 77° F. for 24 hours in non-operating condition, the frequency multiplier, when tested while exposed to the same temperature and relative humidity, shall comply with the requirements of 3.4 for eight hours of continuous operation.

**4.7.1.3** *Altitude.* When tested while exposed to a reduced pressure of 20.58 inches of mercury (equal to an altitude of 10,000 feet), the frequency multiplier shall comply with the requirements of 3.4 for four hours of continuous operation.

**4.7.2** *Non-operating Tests.*

**4.7.2.1** *Temperature.* After having been stored for four hours while exposed and



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

thermally stabilized at an ambient temperature of minus 25° F., the frequency multiplier shall be returned to normal conditions (plus 77° F., relative humidity 60 percent, sea level) and tested for compliance with 3.4. The test shall be repeated at an ambient temperature of plus 158° F.

4.7.2.2 *Altitude.* After having been stored for one hour while exposed to a reduced pressure of 3.44 inches of mercury (equal to an altitude of 50,000 feet), the frequency multiplier shall be returned to normal conditions and tested for compliance with 3.4.

4.7.2.3 *Bench Handling.* On a hardwood table, in its normal operating position, the frequency multiplier shall be lifted on one edge 30° from the horizontal or a maximum of four inches, whichever is less, and dropped. The test shall be repeated for the three other edges for a total of four drops. At the conclusion of the test, the frequency multiplier shall show no physical damage and shall comply with 3.4.

4.7.2.4 *Vibration.* The frequency multiplier shall be securely packed in its commercial shipping container and subjected to a vibration whose frequency shall be varied linearly over the range from 5 to 52 cps with a double amplitude of 0.036 inches. The range shall be traversed from minimum to maximum frequency and returned to minimum frequency within one minute. The test shall be conducted for 15 minutes along each of three mutually perpendicular axes for a total of 45 minutes. At the conclusion of the test, the frequency multiplier shall show no physical damage and shall comply with 3.4.

4.7.2.5 *Shock.* The frequency multiplier shall be securely packed in its commercial container and dropped from a height of three feet on the bottom face, a bottom edge, and a bottom corner. At the conclusion of the test, the frequency multiplier shall show no physical damage and shall comply with 3.4.

4.8 **Sampling.**

4.8.1 *Environmental Tests.* The environmental tests of 4.7 shall be performed on a representative frequency multiplier as specified by the contracting officer.

4.8.2 *Electrical Tests.* The electrical tests of 4.6 shall be performed on each frequency multiplier.

## 5. PREPARATION FOR DELIVERY

**5.1 Packaging.** The frequency multipliers shall be packaged, packed and marked in accordance with normal commercial preservation methods and shall be prepared for delivery to insure acceptance by common or other carrier for safe transportation to point of delivery. The preparation for delivery shall conform to the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment.

## 6. NOTES

**6.1 Intended Use.** The frequency multipliers covered in this specification are components of Microwave Calibration Test Sets AN/USM–129, AN/USM–129A and AN/USM–129B. They are part of the secondary frequency standard equipment used in conjunction with the Universal Microwave Calibrator panels specified in MIL–U–46351 (ORD) when measuring the relative frequency response of passive devices.

**6.2 Ordering Data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable stock number.
(c) Provisions for initial production sample, including date of delivery (see 3.1).
(d) Environmental tests required (see 4.8.1).

**6.3 Design Information.** Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

**Notice.** When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government pro-



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

curement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

**Custodian**
**Army—Ord**

**Preparing Activity:**
**Army—Ord**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.